INCH-POUND

MIL-P-70525 (AR)
AMENDMENT 5
22 July 1996
SUPERSEDING
AMENDMENT 4
18 March 1993

MILITARY SPECIFICATION

PROJECTILE, 105MM, HE, RA, XM913/XM927
MOTOR BODY BONDING ASSEMBLY FOR

This amendment forms a part of Military Specification MIL-P-70525 (AR), dated 29 October 1990, and is approved for use by the U.S. Army Armaments, Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title: Change "XM913", to "M913" (and wherever it appears).
Change "XM927" to "M927" (and wherever it appears).
Change "Projectile" to "Cartridge".

* 1.1 Change "(See 6.11)" on line 4 to read "(See 6.1)"

PAGE 4

3.2 Delete "9391136" and substitute "9381136".

3.3.2 Capitalize "there".

3.3.3 Change "Cracks" to "Discontinuities".

3.3.5.1 Delete "Cracks" in two places and substitute "Discontinuities".

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3.3.5.3 Delete "MIL-R-12562" and substitute "MIL-R-13562".

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* 3.7 Delete in its entirety and replace with the following:

"3.7 Proving ground test. There shall be no motor body bonding assembly related premature explosive function, collapse, separation or break-up of metal parts in the gun bore or in flight. Testing shall be as specified in 4.5.8."

3.8.2 Delete "(See Figure 1)" and substitute "(See Figure 3)".

PAGE 10

Add New Figure 3. X-ray nozzle (included).

PAGE 12

* 4.3.1.1 Change "4.3.2" in line 3 to "4.3.1.2"
Change "Finished Machined Motor Bodies, Unpainted and Unknurled" to "Motor body, Prior to Protective Coating".

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TABLE I. Finished Machined Motor Bodies Unpainted and Unknurled, Mechanical properties, Delete "4.4.3.2/4.5.5" and substitute "4.4.3.1/4.5.7".

Nozzle Inserts, Compression test, Delete "3.6.1" and substitute "3.8.1".

Radiographic Inspection, Delete "3.6.2" and substitute "3.8.2".

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4.4.2.1 Title, delete in its entirety and substitute: "Motor Body, Prior to Weld".

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X-RAY INSPECTION ZONES

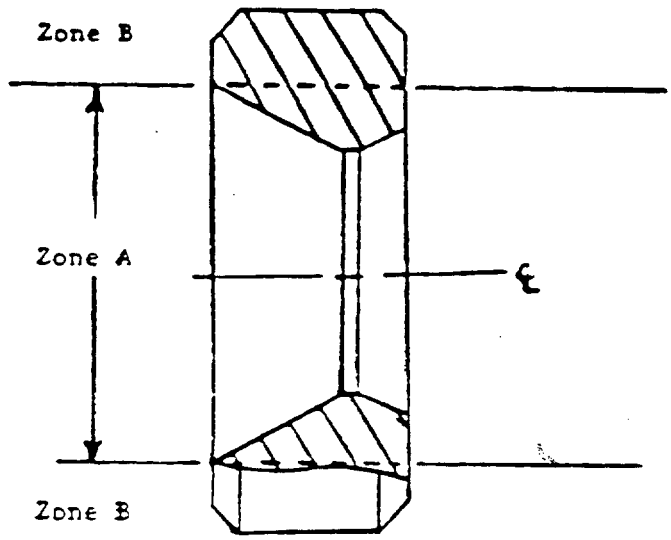


Figure 3. x-ray of nozzle.

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4.4.2.2 Delete title and substitute "Motor Body, protective coating ".

Add the following Special defects:

"2 Ultrasonic C-Scan test	100%	3.6	4.5.4
3 Ultrasonic C-Scan test stamp missing	100%	3.2	Visual"

Major 107 add "(if applicable)".

PAGE 20

4.4.2.2 Major 134, Delete in its entirety and add new Minor 211:
"211 Radius at cavity base Level V 3.2 4.4.3.7/Gage".

* 4.4.2.2, Delete Majors 142 and 143 in their entirety and substitute the following:
"142, Mechanical properties of body, 4.4.3.1.2, 3.2, 4.5.7
143, Properties of rotating band weld, 4.4.3.2, 3.2, 4.5.2"

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Delete Paragraph "4.4.2.2, and substitute "4.4.2.3".

Majors 102 and 103, delete in their entirety and substitute the following:

"102 Salt spray test	4.4.3.6.1	3.2	4.5.5.1
103 Protective coating and thickness	4.4.3.6.2	3.2	4.5.5.2"

* Major 104, Change "3.10" to "3.2".

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* 4.4.2.4 Majors 101, 102, 103 and Minor 201, delete in their entirety and substitute the following:

"101 Radiographic inspection	100%	3.8.2	4.5.3.2
102 Grain flow	4.4.3.5.2	3.2	4.4.3.5.2
103 Compression test	4.4.3.5.1	3.8.1	4.5.3.1
201 True position of throat diameter with outside diameter	Level V	3.2	Gage".

Major 105, delete "(2 Places)".

Add Major 111: "Angle, aft end Level III 3.2 Gage".

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4.4.2.5 Major 101, Delete in its entirety and replace with:

"Adhesive not present on forward O.D. nozzle edge Or rear nozzle I.D. edge (360 degrees coverage required)	100%	3.2	Visual"

Major 102, delete in its entirety and replace with:

"Gap between aft end of nozzle and motor body	Level III	3.2	Gage"

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4.4.3.2.1 Delete the second and third sentence ("The chips for the sample..... selected and dissolved.) and substitute:

"The chips for the sample shall consist of the copper overlay between 4.20-.01 and 4.223-.01 inches diameter (full width). All of the chips shall be collected, crushed, and dissolved (see 6.8.1)."

In the sixth sentence, delete "samplie" and substitute "sample".

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* 4.4.3.8.1 Delete "4.5.9" and substitute "4.5.8". Change "3.10" to "3.7".

4.4.3.8.2 Before last sentence add: "If the lot is rejected, the acceptance criteria reverts back to the beginning of the paragraph".

* Delete "4.5.9" and substitute "4.5.8" and change "3.10" to "3.7" therein.

PAGE 33

4.5.4.1 Second paragraph, first sentence delete "reviously" and substitute "previously".

PAGE 34

4.5.5.1 Delete "4.4.3.5.1" at end of paragraph and substitute "4.4.3.6.1".

* 4.5.8 Delete in its entirety and replace with new paragraph 4.5.8:

"4.5.8 Proving ground. Ballistic testing shall be performed at a Government proving ground in accordance with the applicable acceptance test procedure. The sample motor body assemblies selected in accordance with 4.4.3.8 shall be live loaded with rocket propellant grain and delay assemblies, then assembled with TNT loaded warhead and conditioned to 145-5 degrees F for 16 hours minimum prior to firing rocket on in a M119 series howitzer using a PXR200 propelling charge conditioned to 70+/-2.5 degrees F. The warheads shall be

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fuzed with M577 series MTSQ fuzes set to 40 seconds selected by the proving ground. Photographic and radar coverage shall be provided to assure compliance with 3.7. Range and deflection shall be recorded for informational purposes only."

(Note paragraph 4.5.8.1 added in amendment 4 has been deleted by amendment 5)

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4.5.9 Delete in its entirety without substitution.

5.1 Add the following:

" with MIL-STD-1169, Level C. Appropriate cushioning and partitions shall be provided to protect the items from contacting each other during shipment. Sufficient fillers must be used to insure a tight pack. Adequate corrosion protection shall be provided, if required."

6.0 NOTES. Add the following parenthetical note:

"(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)"

6.2 Acquisition requirements. Delete in its entirety and substitute the following new paragraph:

"6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)."

6.3 Delete in its entirety and replace with new:

"6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract

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Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirements for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.4.4	OT-90-12009	AIE Design Documentation	None

The above DID's were those cleared as of the date of this, specification. The current issue of DOD 5010.12L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

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* 6.4 Delete in its entirety and replace with new:

"6.4 Submission of inspection equipment for design approvals. Gage designs as required shall be submitted to: M913/M927 QUAL. ENGR. (AMSTA-AR-QAA-R), US ARMY ARDEC, PICATINNY ARSENAL, NJ 07806-5000. This address shall be specified on the Contract Data Requirement List, DD Form 1423 in the contract."

* 6.7 Change AMSMC-QAA-Q" to AMSTA-AR-QAA-R" and change "SMCAR-FSA-IM" to "AMSTA-AR-FSA-M".

6.8.1 Add the following paragraph:

"6.8.1 Chemical analysis. An equivalent method may be submitted to the Government for approval."

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes, (additions, modifications, corrections, deletions), from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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